

Bridge Test and Redesign Evidence

NAME _____ DATE _____ PERIOD _____

Safety and Measurement Gate

- ☐ secured supports and catch tray
- ☐ marked 12-inch span and load point
- ☐ known lightweight load stages and safe cap
- ☐ objective stop mark/rule
- ☐ teacher-only load placement
- ☐ everyone behind the marked line
- ☐ no hands, faces, or feet in the test zone
- ☐ dropped weights collected before the next test

If every box is not ready, use the fixed dataset. Never use textbooks or unstable desks.

Shared Result Fields

Code: _____ Route: PHYSICAL / FIXED DATA

Maximum completed safe load stage: _____

Stop reason: _____

First failed/limited element: _____

Physical Route Log

STAGE	KNOWN LOAD	OBSERVATION	CONTINUE / STOP
1			
2			
3			
4			
5 / cap			

One result limitation:

Equal Fixed Dataset

All values below are fictional classroom evidence using the same staged load and stop rule.

SAMPLE	VISIBLE SYSTEM	MAXIMUM COMPLETED STAGE	STOP EVIDENCE
A	Flat beam; two main rails; taped card deck	2	Center deck reached displacement mark before Stage 3
B	Triangulated side frames; cross-braced deck	5 safe cap	No detached joint or displacement stop at cap
C	Curved/segmented sides; narrow support contact	3	One support contact slipped before Stage 4

Which comparison is fair, and what variable is still uncontrolled?

Individual Failure Analysis and Redesign

Result evidence I used:

One specific redesign:

Why that change should affect the result:

What I would measure in the next test:

Career Role and Real-World Limits

Worker who would review this result next:

Additional evidence a real project would require:

One limit of the classroom prototype:
